

EXAMPLE 6 In any triangle ABC, prove that:

$$(i) \frac{a \sin (B-C)}{b^2 - c^2} = \frac{b \sin (C-A)}{c^2 - a^2} = \frac{c \sin (A-B)}{a^2 - b^2}$$

$$(ii) \frac{b^2 - c^2}{\cos B + \cos C} + \frac{c^2 - a^2}{\cos C + \cos A} + \frac{a^2 - b^2}{\cos A + \cos B} = 0$$

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A, b = k \sin B, c = k \sin C$

$$(i) \frac{a \sin (B-C)}{b^2 - c^2} = \frac{k \sin A \sin (B-C)}{k^2 \sin^2 B - k^2 \sin^2 C} = \frac{k \sin (B+C) \sin (B-C)}{k^2 (\sin^2 B - \sin^2 C)}$$

$$= \frac{k (\sin^2 B - \sin^2 C)}{k^2 (\sin^2 B - \sin^2 C)} = \frac{1}{k}$$

and, $\frac{b \sin (C-A)}{c^2 - a^2} = \frac{k \sin B \sin (C-A)}{k^2 \sin^2 C - k^2 \sin^2 A} = \frac{k \sin (C+A) \sin (C-A)}{k^2 (\sin^2 C - \sin^2 A)}$

$$= \frac{k (\sin^2 C - \sin^2 A)}{k^2 (\sin^2 C - \sin^2 A)} = \frac{1}{k}$$

Similarly, it can be shown that $\frac{c \sin (A-B)}{a^2 - b^2} = \frac{1}{k}$

Hence, $\frac{a \sin (B-C)}{b^2 - c^2} = \frac{b \sin (C-A)}{c^2 - a^2} = \frac{c \sin (A-B)}{a^2 - b^2}$

$$(ii) \text{ LHS} = \frac{k^2 (\sin^2 B - \sin^2 C)}{\cos B + \cos C} + \frac{k^2 (\sin^2 C - \sin^2 A)}{\cos C + \cos A} + \frac{k^2 (\sin^2 A - \sin^2 B)}{\cos A + \cos B} \quad [\text{Using (i)}]$$

$$= \frac{k^2 \{(1 - \cos^2 B) - (1 - \cos^2 C)\}}{\cos B + \cos C} + \frac{k^2 \{(1 - \cos^2 C) - (1 - \cos^2 A)\}}{\cos C + \cos A}$$

$$+ \frac{k^2 \{(1 - \cos^2 A) - (1 - \cos^2 B)\}}{\cos A + \cos B}$$

$$= k^2 \left\{ \frac{(\cos^2 C - \cos^2 B)}{\cos B + \cos C} + \frac{(\cos^2 A - \cos^2 C)}{\cos C + \cos A} + \frac{(\cos^2 B - \cos^2 A)}{\cos A + \cos B} \right\}$$

$$= k^2 \{(\cos C - \cos B) + (\cos A - \cos C) + (\cos B - \cos A)\} = k \times 0 = 0 = \text{RHS}$$

EXAMPLE 7 In any triangle ABC, prove that:

$$(i) \frac{1 + \cos (A-B) \cos C}{1 + \cos (A-C) \cos B} = \frac{a^2 + b^2}{a^2 + c^2}$$

$$(ii) a \cos A + b \cos B + c \cos C = 2a \sin B \sin C$$

[NCERT]

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A, b = k \sin B, c = k \sin C$

$$(i) \text{ LHS} = \frac{1 + \cos (A-B) \cos C}{1 + \cos (A-C) \cos B} = \frac{1 - \cos (A-B) \cos (A+B)}{1 - \cos (A-C) \cos (A+C)} = \frac{1 - (\cos^2 A - \sin^2 B)}{1 - (\cos^2 A - \sin^2 C)}$$

$$= \frac{1 - \cos^2 A + \sin^2 B}{1 - \cos^2 A + \sin^2 C} = \frac{\sin^2 A + \sin^2 B}{\sin^2 A + \sin^2 C} = \frac{a^2/k^2 + b^2/k^2}{a^2/k^2 + c^2/k^2} = \frac{a^2 + b^2}{a^2 + c^2} = \text{RHS}$$

$$\begin{aligned}
 \text{(ii)} \quad \text{LHS} &= a \cos A + b \cos B + c \cos C \\
 &= k \sin A \cos A + k \sin B \cos B + k \sin C \cos C \\
 &= \frac{k}{2} \{\sin 2A + \sin 2B + \sin 2C\} = \frac{k}{2} (4 \sin A \sin B \sin C) \\
 &= 2k \sin A \sin B \sin C = 2a \sin B \sin C = \text{RHS} \quad [\because k \sin A = a]
 \end{aligned}$$

EXAMPLE 8 In any triangle ABC, prove that:

$$\text{(i)} \quad \sin \left(\frac{B-C}{2} \right) = \left(\frac{b-c}{a} \right) \cos \frac{A}{2} \quad [\text{NCERT}] \quad \text{(ii)} \quad a \cos \left(\frac{B-C}{2} \right) = (b+c) \sin \frac{A}{2} \quad [\text{NCERT}]$$

$$\text{(iii)} \quad \frac{b-c}{b+c} = \frac{\tan \left(\frac{B-C}{2} \right)}{\tan \left(\frac{B+C}{2} \right)}$$

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

$$\begin{aligned}
 \text{(i)} \quad \text{RHS} &= \left(\frac{b-c}{a} \right) \cos \frac{A}{2} = \left\{ \frac{k \sin B - k \sin C}{k \sin A} \right\} \cos \frac{A}{2} = \left\{ \frac{\sin B - \sin C}{\sin A} \right\} \cos \frac{A}{2} \\
 &= \frac{2 \sin \left(\frac{B-C}{2} \right) \cos \left(\frac{B+C}{2} \right)}{2 \sin \frac{A}{2} \cos \frac{A}{2}} \cos \frac{A}{2} = \frac{\sin \left(\frac{B-C}{2} \right) \cos \left(\frac{\pi-A}{2} \right)}{\sin \frac{A}{2}} \\
 &= \frac{\sin \left(\frac{B-C}{2} \right) \cos \left(\frac{\pi}{2} - \frac{A}{2} \right)}{\sin \frac{A}{2}} = \frac{\sin \left(\frac{B-C}{2} \right) \sin \frac{A}{2}}{\sin \frac{A}{2}} = \sin \left(\frac{B-C}{2} \right) = \text{LHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \text{RHS} &= (b+c) \sin \frac{A}{2} = (k \sin B + k \sin C) \sin \frac{A}{2} = k (\sin B + \sin C) \sin \frac{A}{2} \\
 &= k \times 2 \sin \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right) \sin \frac{A}{2} = 2k \sin \left(\frac{\pi}{2} - \frac{A}{2} \right) \cos \left(\frac{B-C}{2} \right) \sin \frac{A}{2} \\
 &= 2k \cos \frac{A}{2} \cos \left(\frac{B-C}{2} \right) \sin \frac{A}{2} = k \left(2 \sin \frac{A}{2} \cos \frac{A}{2} \right) \cos \left(\frac{B-C}{2} \right) \\
 &= k \sin A \cos \left(\frac{B-C}{2} \right) = a \cos \left(\frac{B-C}{2} \right) = \text{LHS} \quad [\because k \sin A = a]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \text{LHS} &= \frac{b-c}{b+c} = \frac{k \sin B - k \sin C}{k \sin B + k \sin C} = \frac{\sin B - \sin C}{\sin B + \sin C} \\
 &= \frac{2 \sin \left(\frac{B-C}{2} \right) \cos \left(\frac{B+C}{2} \right)}{2 \sin \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right)} = \frac{\tan \left(\frac{B-C}{2} \right)}{\tan \left(\frac{B+C}{2} \right)} = \text{RHS}
 \end{aligned}$$

EXAMPLE 9 In a triangle ABC, if $a \cos A = b \cos B$, show that the triangle is either isosceles or right angled.

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

Now, $a \cos A = b \cos B$

$$\begin{aligned}
 \Rightarrow k \sin A \cos A &= k \sin B \cos B \\
 \Rightarrow 2 \sin A \cos A &= 2 \sin B \cos B \\
 \Rightarrow \sin 2A &= \sin 2B \\
 \Rightarrow 2A &= 2B \text{ or, } 2A = \pi - 2B \\
 \Rightarrow A &= B \text{ or, } A + B = \pi/2 \\
 \Rightarrow A = B \text{ or, } C &= \frac{\pi}{2} \\
 \Rightarrow BC = CA \text{ or, } C &= \frac{\pi}{2} \\
 \Rightarrow \triangle ABC &\text{ is either isosceles or right angled.}
 \end{aligned}$$

EXAMPLE 10 If in a $\triangle ABC$, $\frac{\sin A}{\sin C} = \frac{\sin (A-B)}{\sin (B-C)}$, prove that a^2, b^2, c^2 are in A.P.

SOLUTION Let $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$. Then, $\sin A = ak$, $\sin B = bk$, $\sin C = ck$

Now, $\frac{\sin A}{\sin C} = \frac{\sin (A-B)}{\sin (B-C)}$

$$\Rightarrow \frac{\sin (B+C)}{\sin (A+B)} = \frac{\sin (A-B)}{\sin (B-C)} \quad [\because \sin A = \sin (B+C) \text{ and } \sin C = \sin (A+B)]$$

$$\Rightarrow \sin (B+C) \sin (B-C) = \sin (A+B) \sin (A-B)$$

$$\Rightarrow \sin^2 B - \sin^2 C = \sin^2 A - \sin^2 B$$

$$\Rightarrow k^2 b^2 - k^2 c^2 = k^2 a^2 - k^2 b^2$$

$$\Rightarrow b^2 - c^2 = a^2 - b^2 \Rightarrow 2b^2 = a^2 + c^2 \Rightarrow a^2, b^2, c^2 \text{ are in A.P.}$$

Type II APPLICATIONS OF SINE FORMULA IN PROBLEMS ON HEIGHTS AND DISTANCES

EXAMPLE 11 A tree stands vertically on a hill side which makes an angle of 15° with the horizontal. From a point on the ground 35 m down the hill from the base of the tree, the angle of elevation of the top of the tree is 60° . Find the height of the tree. [NCERT]

SOLUTION Let PQ be the tree on the hill which makes an angle of 15° with the horizontal AR , where A is a point on the ground 35 m down the hill from the base P of the tree.

In $\triangle ARQ$, we have

$$\angle RAQ = 60^\circ \text{ and } \angle ARQ = 90^\circ$$

$$\therefore \angle AQP = 30^\circ$$

In $\triangle APQ$, we have

$$\angle PAQ = 45^\circ \text{ and } \angle AQP = 30^\circ$$

Using Sine rule in $\triangle APQ$, we get

$$\frac{AP}{\sin \angle AQP} = \frac{PQ}{\sin \angle PAQ}$$

$$\Rightarrow \frac{35}{\sin 30^\circ} = \frac{PQ}{\sin 45^\circ} \Rightarrow \frac{35}{1/2} = \frac{PQ}{1/\sqrt{2}} \Rightarrow PQ = \frac{70}{\sqrt{2}} \Rightarrow PQ = 35\sqrt{2} \text{ m.}$$

Hence, height of the tree is $35\sqrt{2}$ m.

EXAMPLE 12 A person, standing on the bank of a river, observes that the angle subtended by a tree on the opposite bank is 60° , when he retreats 20 m from the bank, he finds the angle to be 30° . Find the height of the tree and the breadth of the river.

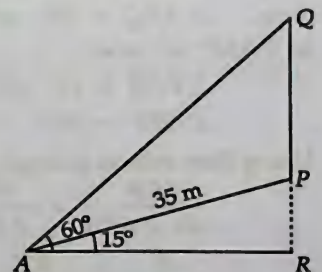


Fig. 10.4

SOLUTION Let AB be the tree at one bank of the river and let the P be the position of the person at the other bank of the river. After retreating 20 m from the bank, let the man be at Q . It is given that $\angle APB = 60^\circ$ and $\angle AQB = 30^\circ$.

Now, $\angle APB = 60^\circ \Rightarrow \angle BPQ = 120^\circ$ and $\angle PBA = 30^\circ$

In ΔBPQ , we have

$$\angle PQB = 30^\circ, \angle BPQ = 120^\circ \text{ and } \angle PBQ = 30^\circ$$

Using Sine rule in triangles BPQ and PAB , we get

$$\frac{BP}{\sin \angle PQB} = \frac{PQ}{\sin \angle PBQ} \text{ and } \frac{BP}{\sin \angle PAB} = \frac{AP}{\sin \angle ABP} = \frac{AB}{\sin \angle APB}$$

$$\Rightarrow \frac{BP}{\sin 30^\circ} = \frac{PQ}{\sin 30^\circ} \text{ and } \frac{BP}{\sin 90^\circ} = \frac{AP}{\sin 30^\circ} = \frac{AB}{\sin 60^\circ}$$

$$\Rightarrow \frac{BP}{1/2} = \frac{20}{1/2} \text{ and } BP = 2AP = \frac{2AB}{\sqrt{3}}$$

$$\Rightarrow BP = 20 \text{ and } BP = 2AP = \frac{2AB}{\sqrt{3}} \quad [\because PQ = 20 \text{ m}]$$

$$\Rightarrow 20 = 2AP = \frac{2AB}{\sqrt{3}}$$

$$\Rightarrow AP = 10 \text{ and } AB = 10\sqrt{3}$$

Hence, the breadth of the river is 10 m and height of the tree is $10\sqrt{3}$ m.

EXAMPLE 13 The angle of elevation of the top point P of the vertical tower PQ of height h from a point A is 45° and from a point B , the angle of elevation is 60° , where B is a point at a distance d from the point A measured along the line AB which makes an angle 30° with AQ . Prove that $d = (\sqrt{3} - 1)h$. [NCERT]

SOLUTION It is given that $\angle PAQ = 45^\circ$ and $\angle BAQ = 30^\circ$. Therefore, $\angle BAP = 15^\circ$.

In ΔAQP , we have

$$\angle PAQ = 45^\circ \text{ and } \angle PQA = 90^\circ$$

$$\therefore \angle APQ = 45^\circ$$

In ΔBRP , we have

$$\angle PBR = 60^\circ \text{ and } \angle PRB = 90^\circ$$

$$\therefore \angle BPR = 30^\circ$$

Now, $\angle APQ = 45^\circ$ and $\angle BPR = 30^\circ \Rightarrow \angle BPA = 15^\circ$

In ΔABP , we have

$$\angle PAB = 15^\circ \text{ and } \angle BPA = 15^\circ.$$

$$\therefore \angle ABP = 150^\circ$$

Using Sine rule in triangle ABP , we get

$$\frac{AB}{\sin \angle APB} = \frac{BP}{\sin \angle PAB} = \frac{AP}{\sin \angle ABP}$$

$$\Rightarrow \frac{d}{\sin 15^\circ} = \frac{BP}{\sin 15^\circ} = \frac{AP}{\sin 150^\circ}$$

$$\Rightarrow \frac{d}{\frac{\sqrt{3}-1}{2\sqrt{2}}} = \frac{AP}{\frac{1}{2}} \Rightarrow AP = \frac{\sqrt{2}d}{\sqrt{3}-1} \quad \dots(i)$$

Using Sine rule in ΔAQP , we get

$$\frac{AP}{\sin \angle AQP} = \frac{AQ}{\sin \angle APQ} = \frac{PQ}{\sin \angle PAQ}$$

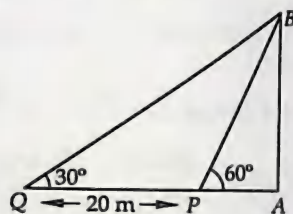


Fig. 10.5

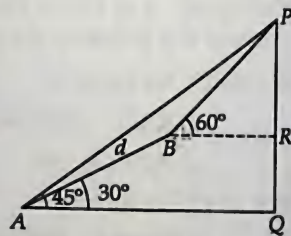


Fig. 10.6

$$\Rightarrow \frac{AP}{\sin 90^\circ} = \frac{PQ}{\sin 45^\circ} \Rightarrow AP = \sqrt{2} PQ \Rightarrow AP = \sqrt{2} h \quad \dots(ii)$$

From (i) and (ii), we get

$$\sqrt{2}h = \frac{\sqrt{2}d}{\sqrt{3}-1} \Rightarrow d = (\sqrt{3}-1)h$$

LEVEL-2

EXAMPLE 14 If in a ΔABC , $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A - B)}{\sin(A + B)}$, prove that it is either a right angled or an isosceles triangle.

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$, (say). Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

$$\begin{aligned} \therefore \frac{a^2 - b^2}{a^2 + b^2} &= \frac{\sin(A - B)}{\sin(A + B)} \\ \Rightarrow \frac{k^2 \sin^2 A - k^2 \sin^2 B}{k^2 \sin^2 A + k^2 \sin^2 B} &= \frac{\sin(A - B)}{\sin(A + B)} \\ \Rightarrow \frac{\sin(A - B)}{\sin(\pi - C)} &= \frac{\sin(A + B) \sin(A - B)}{\sin^2 A + \sin^2 B} \\ \Rightarrow \frac{\sin(A - B)}{\sin C} &= \frac{\sin(\pi - C) \sin(A - B)}{\sin^2 A + \sin^2 B} \\ \Rightarrow \frac{\sin(A - B)}{\sin C} &= \frac{\sin C \sin(A - B)}{\sin^2 A + \sin^2 B} \\ \Rightarrow \sin(A - B) \left\{ \frac{1}{\sin C} - \frac{\sin C}{\sin^2 A + \sin^2 B} \right\} &= 0 \\ \Rightarrow \text{either } \sin(A - B) = 0 \quad \text{or,} \quad \frac{1}{\sin C} - \frac{\sin C}{\sin^2 A + \sin^2 B} &= 0 \\ \Rightarrow \text{either } A - B = 0 \quad \text{or,} \quad \sin^2 A + \sin^2 B - \sin^2 C &= 0 \\ \Rightarrow \text{either } A = B \quad \text{or,} \quad \frac{a^2}{k^2} + \frac{b^2}{k^2} - \frac{c^2}{k^2} &= 0 \quad [\because a = k \sin A, b = k \sin B, c = k \sin C] \\ \Rightarrow \text{either } A = B \quad \text{or,} \quad a^2 + b^2 &= c^2 \\ \Rightarrow \text{either the triangle is isosceles or it is right angled.} \end{aligned}$$

EXAMPLE 15 In any triangle ABC , prove that:

$$(b - c) \cot \frac{A}{2} + (c - a) \cot \frac{B}{2} + (a - b) \cot \frac{C}{2} = 0$$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= (b - c) \cot \frac{A}{2} + (c - a) \cot \frac{B}{2} + (a - b) \cot \frac{C}{2} \\ &= k(\sin B - \sin C) \cot \frac{A}{2} + k(\sin C - \sin A) \cot \frac{B}{2} + k(\sin A - \sin B) \cot \frac{C}{2} \\ &= k \left[2 \sin \left(\frac{B - C}{2} \right) \cos \left(\frac{B + C}{2} \right) \cot \frac{A}{2} + 2 \sin \left(\frac{C - A}{2} \right) \cos \left(\frac{C + A}{2} \right) \cot \frac{B}{2} \right. \\ &\quad \left. + 2 \sin \left(\frac{A - B}{2} \right) \cos \left(\frac{A + B}{2} \right) \cot \frac{C}{2} \right] \end{aligned}$$

$$\begin{aligned}
 &= k \left[2 \sin \left(\frac{B-C}{2} \right) \sin \frac{A}{2} \cot \frac{A}{2} + 2 \sin \left(\frac{C-A}{2} \right) \sin \frac{B}{2} \cot \frac{B}{2} \right. \\
 &\quad \left. + 2 \sin \left(\frac{A-B}{2} \right) \sin \frac{C}{2} \cot \frac{C}{2} \right] \\
 &= k \left[2 \cos \frac{A}{2} \sin \left(\frac{B-C}{2} \right) + 2 \cos \frac{B}{2} \sin \left(\frac{C-A}{2} \right) + 2 \cos \frac{C}{2} \sin \left(\frac{A-B}{2} \right) \right] \\
 &= 2k \left[\sin \left(\frac{B+C}{2} \right) \sin \left(\frac{B-C}{2} \right) + \sin \left(\frac{C+A}{2} \right) \sin \left(\frac{C-A}{2} \right) + \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right) \right] \\
 &\quad \left[\because \cos \frac{A}{2} = \sin \left(\frac{B+C}{2} \right) \text{ etc.} \right] \\
 &= 2k \left\{ \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} + \sin^2 \frac{C}{2} - \sin^2 \frac{A}{2} + \sin^2 \frac{A}{2} - \sin^2 \frac{B}{2} \right\} = 2k \times 0 = 0 = \text{RHS}
 \end{aligned}$$

EXAMPLE 16 Let O be a point inside a triangle ABC such that $\angle OAB = \angle OBC = \angle OCA = \omega$, then show that:

(i) $\cot \omega = \cot A + \cot B + \cot C$

(ii) $\operatorname{cosec}^2 \omega = \operatorname{cosec}^2 A + \operatorname{cosec}^2 B + \operatorname{cosec}^2 C$

SOLUTION (i) In $\triangle OBC$,

$$\angle OCB = \angle C - \omega \text{ and, } \angle BOC = 180 - \omega - (C - \omega) = 180^\circ - C$$

Similarly, we obtain $\angle AOB = 180^\circ - B$

Applying sine rule in $\triangle OAB$, we obtain

$$\begin{aligned}
 \frac{OB}{\sin \angle OAB} &= \frac{AB}{\sin \angle AOB} \\
 \frac{OB}{\sin \omega} &= \frac{AB}{\sin (180^\circ - B)} \\
 \Rightarrow \frac{OB}{\sin \omega} &= \frac{c}{\sin B} \\
 \Rightarrow OB &= \frac{c \sin \omega}{\sin B} \quad \dots(i)
 \end{aligned}$$

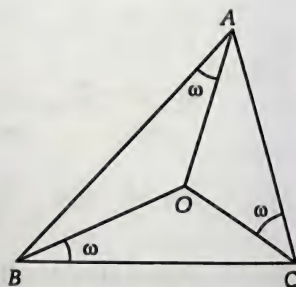


Fig. 10.7

Applying sine rule in $\triangle OBC$, we get

$$\begin{aligned}
 \frac{OB}{\sin \angle BCO} &= \frac{BC}{\sin \angle BOC} \\
 \Rightarrow \frac{OB}{\sin (C - \omega)} &= \frac{BC}{\sin (180^\circ - C)} \Rightarrow \frac{OB}{\sin (C - \omega)} = \frac{a}{\sin C} \Rightarrow OB = \frac{a \sin (C - \omega)}{\sin C} \quad \dots(ii)
 \end{aligned}$$

From (i) and (ii), we get

$$\begin{aligned}
 \frac{c \sin \omega}{\sin B} &= \frac{a \sin (C - \omega)}{\sin C} \\
 \Rightarrow \frac{k \sin C \sin \omega}{\sin B} &= \frac{k \sin A \sin (C - \omega)}{\sin C} \quad [\text{Using sine rule}] \\
 \Rightarrow \sin^2 C \sin \omega &= \sin A \sin B \sin (C - \omega) \\
 \Rightarrow \sin C \sin (A + B) \sin \omega &= \sin A \sin B \sin (C - \omega) \quad [\because \sin C = \sin (\pi - (A + B)) = \sin (A + B)] \\
 \Rightarrow \frac{\sin (A + B)}{\sin A \sin B} &= \frac{\sin (C - \omega)}{\sin C \sin \omega} \\
 \Rightarrow \frac{\sin A \cos B + \cos A \sin B}{\sin A \sin B} &= \frac{\sin C \cos \omega - \cos C \sin \omega}{\sin C \sin \omega} \\
 \Rightarrow \cot B + \cot A &= \cot \omega - \cot C \Rightarrow \cot \omega = \cot A + \cot B + \cot C
 \end{aligned}$$

(ii) From (i), we have

$$\begin{aligned}\cot \omega &= \cot A + \cot B + \cot C \\ \Rightarrow \cot^2 \omega &= \cot^2 A + \cot^2 B + \cot^2 C + 2(\cot A \cot B + \cot B \cot C + \cot C \cot A) \\ \Rightarrow \cot^2 \omega &= \cot^2 A + \cot^2 B + \cot^2 C + 2 \quad [\because \cot A \cot B + \cot B \cot C + \cot C \cot A = 1] \\ \Rightarrow \operatorname{cosec}^2 \omega - 1 &= (\operatorname{cosec}^2 A - 1) + (\operatorname{cosec}^2 B - 1) + (\operatorname{cosec}^2 C - 1) + 2 \\ \Rightarrow \operatorname{cosec}^2 \omega &= \operatorname{cosec}^2 A + \operatorname{cosec}^2 B + \operatorname{cosec}^2 C\end{aligned}$$

EXAMPLE 17 The angle of elevation of the top of a tower from a point A due South of the tower is α and from B due East of the tower is β . If $AB = d$, show that the height of the tower is $\frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta}}$.

SOLUTION Let OP be the tower and let A and B be two points due South and East respectively of the tower such that $\angle OAP = \alpha$ and $\angle OBP = \beta$. Then,

$$\angle OPA = \frac{\pi}{2} - \alpha \text{ and } \angle OPB = \frac{\pi}{2} - \beta.$$

Using sine rule in Δ 's OAP and OBP , we have

$$\frac{OA}{\sin\left(\frac{\pi}{2} - \alpha\right)} = \frac{OP}{\sin \alpha} \text{ and } \frac{OB}{\sin\left(\frac{\pi}{2} - \beta\right)} = \frac{OP}{\sin \beta}$$

$$\Rightarrow \frac{OA}{\cos \alpha} = \frac{OP}{\sin \alpha} \text{ and } \frac{OB}{\cos \beta} = \frac{OP}{\sin \beta}$$

$$\Rightarrow OA = OP \cot \alpha \text{ and } OB = OP \cot \beta$$

Using Pythagoras theorem in ΔAOB , we get

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow d^2 = OP^2 \cot^2 \alpha + OP^2 \cot^2 \beta$$

$$\Rightarrow OP = \frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta}}$$

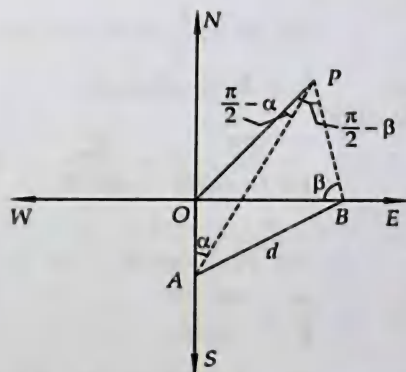


Fig. 10.8

EXAMPLE 18 The elevation of a tower at a station A due North of it is α and at a station B due West of A is β . Prove that the height of the tower is $\frac{AB \sin \alpha \sin \beta}{\sqrt{\sin^2 \alpha - \sin^2 \beta}}$.

SOLUTION Let OP be the tower and let A be a point due north of the tower OP and let B be the point due west of A such that $\angle OAP = \alpha$ and $\angle OBP = \beta$.

Clearly, triangles AOP and BOP are right triangles right angled at O.

$$\therefore \angle OPA = \frac{\pi}{2} - \alpha \text{ and } \angle OPB = \frac{\pi}{2} - \beta$$

Using sine rule in triangles AOP and BOP , we get

$$\frac{OA}{\sin\left(\frac{\pi}{2} - \alpha\right)} = \frac{OP}{\sin \alpha} \text{ and } \frac{OB}{\sin\left(\frac{\pi}{2} - \beta\right)} = \frac{OP}{\sin \beta}$$

$$\Rightarrow OA = OP \cot \alpha \text{ and } OB = OP \cot \beta$$

Applying Pythagoras theorem in ΔOAB , we get

$$OB^2 = OA^2 + AB^2$$

$$\Rightarrow OB^2 - OA^2 = AB^2$$

$$\Rightarrow OP^2 \cot^2 \beta - OP^2 \cot^2 \alpha = AB^2$$

$$\Rightarrow OP^2 (\cot^2 \beta - \cot^2 \alpha) = AB^2$$

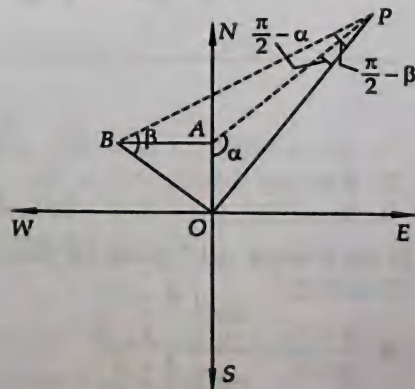


Fig. 10.9

$$\Rightarrow OP^2 (\operatorname{cosec}^2 \beta - \operatorname{cosec}^2 \alpha) = AB^2$$

$$\Rightarrow OP^2 \frac{(\sin^2 \alpha - \sin^2 \beta)}{\sin^2 \alpha \sin^2 \beta} = AB^2 \Rightarrow OP = \frac{AB \sin \alpha \sin \beta}{\sqrt{\sin^2 \alpha - \sin^2 \beta}}$$

EXAMPLE 19 An object is observed from three points A, B, C in the same horizontal line passing through the base of the object. The angle of elevation at B is twice and at C thrice that at A. If $AB = a$, $BC = b$ prove that the height of the object is $\frac{a}{2b} \sqrt{(a+b)(3b-a)}$.

SOLUTION Let the object be at P at a height h from OA. Let the object when observed from A, B and C the angles of elevation are θ , 2θ and 3θ respectively.

In $\triangle PAB$, we have

$$2\theta = \theta + \angle APB \Rightarrow \angle APB = \theta$$

$$\therefore \angle PAB = \angle APB = \theta \Rightarrow AB = BP = a$$

Similarly, in triangle BPC , $\angle BPC = \theta$.

In $\triangle OPB$

$$\sin 2\theta = \frac{h}{a} \Rightarrow h = a \sin 2\theta$$

$$\Rightarrow h = 2a \sin \theta \cos \theta \quad \dots(i)$$

In $\triangle PBC$

$$\frac{PB}{\sin(\pi - 3\theta)} = \frac{BC}{\sin \theta} \quad [\text{Using sine rule}]$$

$$\Rightarrow \frac{a}{\sin 3\theta} = \frac{b}{\sin \theta}$$

$$\Rightarrow \frac{a}{b} = \frac{\sin 3\theta}{\sin \theta}$$

$$\Rightarrow \frac{a}{b} = \frac{3 \sin \theta - 4 \sin^3 \theta}{\sin \theta}$$

$$\Rightarrow \frac{a}{b} = 3 - 4 \sin^2 \theta \Rightarrow 4 \sin^2 \theta = 3 - \frac{a}{b} \Rightarrow \sin^2 \theta = \frac{3b - a}{4b} \Rightarrow \sin \theta = \sqrt{\frac{3b - a}{4b}}$$

$$\therefore \cos^2 \theta = 1 - \sin^2 \theta \Rightarrow \cos^2 \theta = 1 - \frac{3b - a}{4b} = \frac{a + b}{4b} \Rightarrow \cos \theta = \sqrt{\frac{a + b}{4b}}$$

Substituting the values of $\sin \theta$ and $\cos \theta$ in (i), we get

$$h = 2a \sqrt{\frac{3b - a}{4b}} \times \sqrt{\frac{a + b}{4b}} = \frac{a}{2b} \sqrt{(a + b)(3b - a)}$$

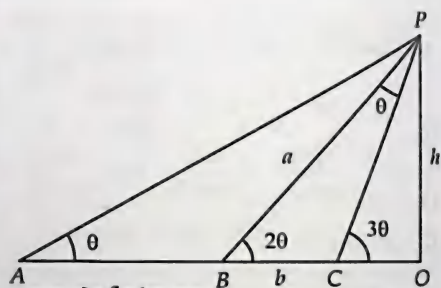


Fig. 10.10

EXERCISE 10.1

LEVEL-1

1. If in a $\triangle ABC$, $\angle A = 45^\circ$, $\angle B = 60^\circ$, and $\angle C = 75^\circ$; find the ratio of its sides.
2. If in any $\triangle ABC$, $\angle C = 105^\circ$, $\angle B = 45^\circ$, $a = 2$, then find b .
3. In $\triangle ABC$, if $a = 18$, $b = 24$ and $C = 30$ and $\angle C = 90^\circ$, find $\sin A$, $\sin B$ and $\sin C$.

In any triangle ABC , prove the following: (4-24)

$$4. \frac{a-b}{a+b} = \frac{\tan\left(\frac{A-B}{2}\right)}{\tan\left(\frac{A+B}{2}\right)}$$

$$5. (a-b) \cos \frac{C}{2} = c \sin \left(\frac{A-B}{2} \right)$$

[NCERT]

$$6. \frac{c}{a-b} = \frac{\tan \left(\frac{A}{2} \right) + \tan \left(\frac{B}{2} \right)}{\tan \left(\frac{A}{2} \right) - \tan \left(\frac{B}{2} \right)}$$

$$7. \frac{c}{a+b} = \frac{1 - \tan \left(\frac{A}{2} \right) \tan \left(\frac{B}{2} \right)}{1 + \tan \left(\frac{A}{2} \right) \tan \left(\frac{B}{2} \right)}$$

$$8. \frac{a+b}{c} = \frac{\cos \left(\frac{A-B}{2} \right)}{\sin \frac{C}{2}}$$

[NCERT]

$$9. \sin \left(\frac{B-C}{2} \right) = \frac{b-c}{a} \cos \frac{A}{2}$$

$$10. \frac{a^2 - c^2}{b^2} = \frac{\sin (A-C)}{\sin (A+C)}$$

$$11. b \sin B - c \sin C = a \sin (B-C)$$

$$12. a^2 \sin (B-C) = (b^2 - c^2) \sin A$$

$$13. \frac{\sqrt{\sin A} - \sqrt{\sin B}}{\sqrt{\sin A} + \sqrt{\sin B}} = \frac{a+b-2\sqrt{ab}}{a-b}$$

$$14. a (\sin B - \sin C) + b (\sin C - \sin A) + c (\sin A - \sin B) = 0$$

$$15. \frac{a^2 \sin (B-C)}{\sin A} + \frac{b^2 \sin (C-A)}{\sin B} + \frac{c^2 \sin (A-B)}{\sin C} = 0$$

$$16. a^2 (\cos^2 B - \cos^2 C) + b^2 (\cos^2 C - \cos^2 A) + c^2 (\cos^2 A - \cos^2 B) = 0$$

$$17. b \cos B + c \cos C = a \cos (B-C)$$

$$18. \frac{\cos 2A}{a^2} - \frac{\cos 2B}{b^2} = \frac{1}{a^2} - \frac{1}{b^2}$$

$$19. \frac{\cos^2 B - \cos^2 C}{b+c} + \frac{\cos^2 C - \cos^2 A}{c+a} + \frac{\cos^2 A - \cos^2 B}{a+b} = 0$$

$$20. a \sin \frac{A}{2} \sin \left(\frac{B-C}{2} \right) + b \sin \frac{B}{2} \sin \left(\frac{C-A}{2} \right) + c \sin \frac{C}{2} \sin \left(\frac{A-B}{2} \right) = 0.$$

$$21. \frac{b \sec B + c \sec C}{\tan B + \tan C} = \frac{c \sec C + a \sec A}{\tan C + \tan A} = \frac{a \sec A + b \sec B}{\tan A + \tan B}.$$

$$22. a \cos A + b \cos B + c \cos C = 2b \sin A \sin C = 2c \sin A \sin B$$

$$23. a (\cos B \cos C + \cos A) = b (\cos C \cos A + \cos B) = c (\cos A \cos B + \cos C).$$

$$24. a (\cos C - \cos B) = 2(b-c) \cos^2 \frac{A}{2}.$$

[NCERT]

$$25. \text{In } \triangle ABC \text{ prove that, if } \theta \text{ be any angle, then } b \cos \theta = c \cos (A - \theta) + a \cos (C + \theta).$$

$$26. \text{In a } \triangle ABC, \text{ if } \sin^2 A + \sin^2 B = \sin^2 C, \text{ show that the triangle is right angled.}$$

LEVEL-2

27. In any $\triangle ABC$, if a^2, b^2, c^2 are in A.P., prove that $\cot A, \cot B$ and $\cot C$ are also in A.P.
28. The upper part of a tree broken over by the wind makes an angle of 30° with the ground and the distance from the root to the point where the top of the tree touches the ground is 15 m. Using sine rule, find the height of the tree.
29. At the foot of a mountain the elevation of its summit is 45° ; after ascending 1000 m towards the mountain up a slope of 30° inclination, the elevation is found to be 60° . Find the height of the mountain.
30. A person observes the angle of elevation of the peak of a hill from a station to be α . He walks c metres along a slope inclined at the angle β and finds the angle of elevation of the peak of the hill to be γ . Show that the height of the peak above the ground is $\frac{c \sin \alpha \sin (\gamma - \beta)}{(\sin \gamma - \alpha)}$.
31. If the sides a, b, c of a $\triangle ABC$ are in H.P., prove that $\sin^2 \frac{A}{2}, \sin^2 \frac{B}{2}, \sin^2 \frac{C}{2}$ are in H.P.

ANSWERS

1. $2:\sqrt{6}:\sqrt{3}+1$

2. $2\sqrt{2}$

3. $\sin A = \frac{3}{5}, \sin B = \frac{4}{5}$

28. $15\sqrt{3}$ m

29. $500(\sqrt{3} + 1)$ metres

10.3 THE LAW OF COSINES

THEOREM In any $\triangle ABC$, we have:

$$(i) a^2 = b^2 + c^2 - 2bc \cos A \text{ or, } \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$(ii) b^2 = c^2 + a^2 - 2ac \cos B \text{ or, } \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$(iii) c^2 = a^2 + b^2 - 2ab \cos C \text{ or, } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

PROOF The following cases may arise:

CASE I When $\triangle ABC$ is an acute angled triangle:

Draw perpendicular AD from A on BC .

In $\triangle ABD$, we have

$$\cos B = \frac{BD}{c} \Rightarrow BD = c \cos B \quad \dots(i)$$

In $\triangle ACD$, we have

$$\cos C = \frac{CD}{b} \Rightarrow CD = b \cos C$$

In $\triangle ACD$, using Pythagoras theorem, we have

$$AC^2 = AD^2 + CD^2$$

$$\Rightarrow AC^2 = AD^2 + (BC - BD)^2$$

$$\Rightarrow AC^2 = AD^2 + BC^2 + BD^2 - 2BC \cdot BD$$

$$\Rightarrow AC^2 = BC^2 + (AD^2 + BD^2) - 2BC \cdot BD$$

$$\Rightarrow AC^2 = BC^2 + AB^2 - 2BC \cdot BD$$

$$\Rightarrow b^2 = a^2 + c^2 - 2ac \cos B$$

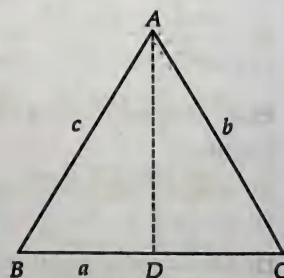


Fig. 10.11

$$[\because AB^2 = BD^2 + AD^2]$$

[Using (i)]

$$\Rightarrow b^2 = c^2 + a^2 - 2ca \cos B$$

$$\Rightarrow \cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

CASE II When $\triangle ABC$ is an obtuse angled triangle:

Draw perpendicular AD from A on CB produced.

In $\triangle ABD$, we have

$$\cos(180 - B) = \frac{BD}{AB} \Rightarrow BD = -AB \cos B = -c \cos B \quad \dots(i)$$

Using Pythagoras theorem in $\triangle ACD$, we have

$$AC^2 = AD^2 + CD^2$$

$$\Rightarrow AC^2 = AD^2 + (CB + BD)^2$$

$$\Rightarrow AC^2 = AD^2 + CB^2 + BD^2 + 2CB \cdot BD$$

$$\Rightarrow AC^2 = BC^2 + (BD^2 + AD^2) + 2BC \cdot BD$$

$$\Rightarrow AC^2 = BC^2 + AB^2 + 2BC \cdot BD$$

$$\Rightarrow b^2 = a^2 + c^2 + 2a(-c \cos B)$$

$$\Rightarrow b^2 = c^2 + a^2 - 2ac \cos B$$

$$\Rightarrow \cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

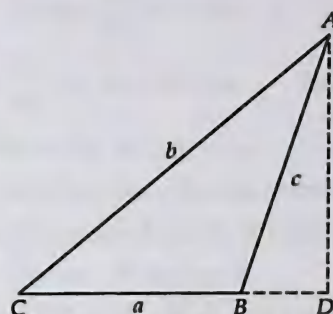


Fig. 10.12

$$[\text{In } \triangle ABD, AB^2 = AD^2 + BD^2]$$

[Using (i)]

CASE III When $\triangle ABC$ is a right angled triangle:

Let $\triangle ABC$ be a right angled triangle with right angle at B . Then, by Pythagoras theorem, we obtain

$$b^2 = a^2 + c^2$$

$$\Rightarrow b^2 = a^2 + c^2 - 2ac \cos B$$

$$\left[\because B = \frac{\pi}{2} \therefore \cos B = 0 \right]$$

$$\Rightarrow \cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

Hence, in all the cases, we have

$$b^2 = c^2 + a^2 - 2ac \cos B \Rightarrow \cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

In a similar manner other results can be proved.

10.4 PROJECTION FORMULAE

THEOREM In any $\triangle ABC$, we have

$$(i) a = b \cos C + c \cos B \quad (ii) b = c \cos A + a \cos C \quad (iii) c = a \cos B + b \cos A$$

i.e. any side of a triangle is equal to the sum of the projections of other two sides on it.

PROOF The following cases arise:

CASE I When $\triangle ABC$ is an acute angled triangle:

In Fig. 10.1, we have

$$\cos B = \frac{BD}{AB} \Rightarrow BD = AB \cos B \Rightarrow BD = c \cos B$$

$$\text{and, } \cos C = \frac{CD}{AC} \Rightarrow CD = AC \cos C \Rightarrow CD = b \cos C$$

$$\text{Hence, } a = BC = BD + CD \Rightarrow a = c \cos B + b \cos C$$

CASE II When $\triangle ABC$ is an obtuse angled triangle

In Fig. 10.2, we have

$$\cos C = \frac{CD}{AC} \Rightarrow CD = AC \cos C \Rightarrow CD = b \cos C$$

$$\text{and, } \cos (180 - B) = \frac{BD}{AB} \Rightarrow BD = AB \cos (180 - B) \Rightarrow BD = -c \cos B$$

$$\therefore a = BC = CD - BD \Rightarrow a = b \cos C + c \cos B$$

Hence, in each case, we obtain $a = b \cos C + c \cos B$

Similarly, it can be proved that

$$b = c \cos A + a \cos C \text{ and } c = a \cos B + b \cos A$$

Q.E.D.

REMARK In Fig. 10.1, BD and CD are the projections of AB and AC respectively on BC .

10.5 NAPIER'S ANALOGY (LAW OF TANGENTS)

THEOREM In any $\triangle ABC$, we have

$$(i) \tan \left(\frac{B-C}{2} \right) = \left(\frac{b-c}{b+c} \right) \cot \frac{A}{2} \quad (ii) \tan \left(\frac{A-B}{2} \right) = \left(\frac{a-b}{a+b} \right) \cot \frac{C}{2}$$

$$(iii) \tan \left(\frac{C-A}{2} \right) = \left(\frac{c-a}{c+a} \right) \cot \frac{B}{2}$$

PROOF Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A, b = k \sin B, c = k \sin C$... (i)

$$(i) \quad \text{RHS} = \frac{b-c}{b+c} \cot \frac{A}{2} = \frac{k \sin B - k \sin C}{k \sin B + k \sin C} \cot \frac{A}{2} \quad [\text{Using (i)}]$$

$$\begin{aligned} &= \left(\frac{\sin B - \sin C}{\sin B + \sin C} \right) \cot \frac{A}{2} = \left[\frac{2 \sin \left(\frac{B-C}{2} \right) \cos \left(\frac{B+C}{2} \right)}{2 \sin \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right)} \right] \cot \frac{A}{2} \\ &= \tan \left(\frac{B-C}{2} \right) \cot \left(\frac{B+C}{2} \right) \cot \frac{A}{2} = \tan \left(\frac{B-C}{2} \right) \cot \left(\frac{\pi}{2} - \frac{A}{2} \right) \cot \frac{A}{2} \\ &= \tan \left(\frac{B-C}{2} \right) \tan \frac{A}{2} \cot \frac{A}{2} = \tan \left(\frac{B-C}{2} \right) = \text{LHS} \end{aligned}$$

Similarly, (ii) and (iii) can be proved.

10.6 AREA OF A TRIANGLE

THEOREM Prove that the area of $\triangle ABC$ is given by

$$\Delta = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B = \frac{1}{2} ab \sin C$$

PROOF Let ABC be a triangle. Then the following cases arise :

CASE I When $\triangle ABC$ is an acute angled triangle:

In Fig. 10.1, we have

$$\sin B = \frac{AD}{AB} \Rightarrow AD = AB \sin B = c \sin B$$

$$\therefore \Delta = \text{Area of } \triangle ABC = \frac{1}{2} BC \times AD = \frac{1}{2} a c \sin B$$

CASE II When $\triangle ABC$ is an obtuse angled triangle:

In Fig. 10.2, we have

$$\sin (180 - B) = \frac{AD}{AB} \Rightarrow AD = AB \sin B = c \sin B$$

$$\therefore \Delta = \text{Area of } \triangle ABC = \frac{1}{2} BC \times AD = \frac{1}{2} a c \sin B$$

Thus, in each case, we have $\Delta = \frac{1}{2} ac \sin B$

Similarly, it can be proved that $\Delta = \frac{1}{2} ab \sin C$ and $\Delta = \frac{1}{2} bc \sin A$

Q.E.D.

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I PROBLEMS ON APPLICATIONS OF COSINE FORMULA AND SINE RULE

EXAMPLE 1 In a $\triangle ABC$, if $a = 3$, $b = 5$ and $c = 7$, find $\cos A$, $\cos B$ and $\cos C$.

SOLUTION We have, $a = 3$, $b = 5$ and $c = 7$

$$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{25 + 49 - 9}{2 \times 5 \times 7} = \frac{65}{70} = \frac{13}{14}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2bc} = \frac{49 + 9 - 25}{2 \times 3 \times 7} = \frac{33}{42} = \frac{11}{14}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{9 + 25 - 49}{2 \times 3 \times 5} = -\frac{15}{30} = -\frac{1}{2}$$

EXAMPLE 2 If the sides of a $\triangle ABC$ are $a = 4$, $b = 6$ and $c = 8$, show that $4 \cos B + 3 \cos C = 2$.

SOLUTION We have, $a = 4$, $b = 6$ and $c = 8$

$$\therefore \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{16 + 64 - 36}{2 \times 4 \times 8} = \frac{44}{64} = \frac{11}{16}$$

$$\text{and, } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{16 + 36 - 64}{2 \times 4 \times 6} = -\frac{12}{48} = -\frac{1}{4}$$

$$\therefore 4 \cos B + 3 \cos C = 4 \times \frac{11}{16} - \frac{3}{4} = \frac{11}{4} - \frac{3}{4} = 2$$

EXAMPLE 3 In any $\triangle ABC$, prove that:

$$(i) a(b \cos C - c \cos B) = b^2 - c^2$$

[NCERT]

$$(ii) \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$

[NCERT]

$$(iii) 2(bc \cos A + ca \cos B + ab \cos C) = a^2 + b^2 + c^2$$

SOLUTION (i) LHS = $a(b \cos C - c \cos B) = ab \cos C - ac \cos B$

$$= ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right) - ac \left(\frac{a^2 + c^2 - b^2}{2ac} \right)$$

$$= \frac{1}{2} \left\{ (a^2 + b^2 - c^2) - (a^2 + c^2 - b^2) \right\} = b^2 - c^2 = \text{RHS}$$

$$\begin{aligned}
 \text{(ii)} \quad \text{LHS} &= \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} \\
 &= \frac{b^2 + c^2 - a^2}{2abc} + \frac{c^2 + a^2 - b^2}{2acb} + \frac{a^2 + b^2 - c^2}{2abc} = \frac{a^2 + b^2 + c^2}{2abc} = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \text{LHS} &= 2bc \cos A + 2ca \cos B + 2ab \cos C \\
 &= 2bc \left(\frac{b^2 + c^2 - a^2}{2bc} \right) + 2ca \left(\frac{c^2 + a^2 - b^2}{2ac} \right) + 2ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \\
 &= (b^2 + c^2 - a^2) + (c^2 + a^2 - b^2) + (a^2 + b^2 - c^2) = a^2 + b^2 + c^2 = \text{RHS}
 \end{aligned}$$

EXAMPLE 4 In any $\triangle ABC$, prove that: $(a-b)^2 \cos^2 \frac{C}{2} + (a+b)^2 \sin^2 \frac{C}{2} = c^2$

PROOF We have,

$$\begin{aligned}
 \text{LHS} &= (a-b)^2 \cos^2 \frac{C}{2} + (a+b)^2 \sin^2 \frac{C}{2} \\
 &= a^2 \left(\cos^2 \frac{C}{2} + \sin^2 \frac{C}{2} \right) + b^2 \left(\cos^2 \frac{C}{2} + \sin^2 \frac{C}{2} \right) - 2ab \left(\cos^2 \frac{C}{2} - \sin^2 \frac{C}{2} \right) \\
 &= a^2 + b^2 - 2ab \cos C = c^2 = \text{RHS}
 \end{aligned}$$

EXAMPLE 5 In a $\triangle ABC$, prove that:

$$(i) (b^2 - c^2) \cot A + (c^2 - a^2) \cot B + (a^2 - b^2) \cot C = 0 \quad [\text{NCERT}]$$

$$(ii) \left(\frac{b^2 - c^2}{a^2} \right) \sin 2A + \left(\frac{c^2 - a^2}{b^2} \right) \sin 2B + \left(\frac{a^2 - b^2}{c^2} \right) \sin 2C = 0 \quad [\text{NCERT}]$$

SOLUTION Let $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$. Then, $\sin A = ak$, $\sin B = bk$ and $\sin C = ck$

$$\begin{aligned}
 (i) \text{ LHS} &= (b^2 - c^2) \cot A + (c^2 - a^2) \cot B + (a^2 - b^2) \cot C \\
 &= (b^2 - c^2) \frac{\cos A}{\sin A} + (c^2 - a^2) \frac{\cos B}{\sin B} + (a^2 - b^2) \frac{\cos C}{\sin C} \\
 &= \frac{(b^2 - c^2)}{ka} \left(\frac{b^2 + c^2 - a^2}{2bc} \right) + \frac{(c^2 - a^2)}{kb} \left(\frac{a^2 + c^2 - b^2}{2ac} \right) + \frac{(a^2 - b^2)}{kc} \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \\
 &= \frac{1}{2kabc} \left\{ (b^2 - c^2)(b^2 + c^2 - a^2) + (c^2 - a^2)(a^2 + c^2 - b^2) + (a^2 - b^2)(a^2 + b^2 - c^2) \right\} \\
 &= \frac{1}{2kabc} \left\{ (b^2 - c^2)(b^2 + c^2) - a^2(b^2 - c^2) + (c^2 - a^2)(c^2 + a^2) - b^2(c^2 - a^2) \right. \\
 &\quad \left. + (a^2 - b^2)(a^2 + b^2) - c^2(a^2 - b^2) \right\} \\
 &= \frac{1}{2kabc} \left\{ (b^2 - c^2)(b^2 + c^2) + (c^2 - a^2)(c^2 + a^2) + (a^2 - b^2)(a^2 + b^2) \right. \\
 &\quad \left. - a^2(b^2 - c^2) - b^2(c^2 - a^2) - c^2(a^2 - b^2) \right\} \\
 &= \frac{1}{2kabc} \left\{ (b^4 - c^4) + (c^4 - a^4) + (a^4 - b^4) - (a^2b^2 - a^2c^2) - (b^2c^2 - b^2a^2) - (c^2a^2 - c^2b^2) \right\} \\
 &= \frac{1}{2kabc} \times 0 = 0 = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) LHS} &= \left(\frac{b^2 - c^2}{a^2} \right) \sin 2A + \left(\frac{c^2 - a^2}{b^2} \right) \sin 2B + \left(\frac{a^2 - b^2}{c^2} \right) \sin 2C \\
 &= \left(\frac{b^2 - c^2}{a^2} \right) 2 \sin A \cos A + \left(\frac{c^2 - a^2}{b^2} \right) 2 \sin B \cos B + \left(\frac{a^2 - b^2}{c^2} \right) 2 \sin C \cos C \\
 &= \left(\frac{b^2 - c^2}{a^2} \right) 2ka \left(\frac{b^2 + c^2 - a^2}{2bc} \right) + \left(\frac{c^2 - a^2}{b^2} \right) 2kb \left(\frac{a^2 + c^2 - b^2}{2ac} \right) + \left(\frac{a^2 - b^2}{c^2} \right) 2kc \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \\
 &= \frac{k}{abc} \left\{ (b^2 - c^2)(b^2 + c^2 - a^2) + (c^2 - a^2)(c^2 + a^2 - b^2) + (a^2 - b^2)(a^2 + b^2 - c^2) \right\} \\
 &= \frac{k}{abc} \times 0 = 0 = \text{RHS}
 \end{aligned}$$

Type II PROBLEMS BASED ON COSINE, SINE AND PROJECTION FORMULAE**EXAMPLE 6** In any $\triangle ABC$, prove that:

$$\text{(i) } \frac{\sin B}{\sin C} = \frac{c - a \cos B}{b - a \cos C}$$

$$\text{(ii) } 2 \left\{ a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2} \right\} = a + c - b$$

$$\text{(iii) } 2 \left\{ b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2} \right\} = a + b + c$$

$$\text{(iv) } (b + c) \cos A + (c + a) \cos B + (a + b) \cos C = a + b + c$$

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

$$\text{(i) RHS} = \frac{c - a \cos B}{b - a \cos C} = \frac{(a \cos B + b \cos A) - a \cos B}{(a \cos C + c \cos A) - a \cos C} = \frac{b \cos A}{c \cos A} = \frac{b}{c} = \frac{k \sin B}{k \sin C} = \frac{\sin B}{\sin C} = \text{LHS}$$

$$\begin{aligned}
 \text{(ii) LHS} &= 2 \left(a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2} \right) = \left\{ a(1 - \cos C) + c(1 - \cos A) \right\} \\
 &= a + c - (a \cos C + c \cos A) = a + c - b = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) LHS} &= 2 \left(b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2} \right) = \left\{ b(1 + \cos C) + c(1 + \cos B) \right\} \\
 &= (b + c + b \cos C + c \cos B) = b + c + a = a + b + c = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) LHS} &= (b + c) \cos A + (c + a) \cos B + (a + b) \cos C \\
 &= (b \cos A + a \cos B) + (c \cos A + a \cos C) + (b \cos C + c \cos B) \\
 &= c + b + a = a + b + c = \text{RHS}
 \end{aligned}$$

EXAMPLE 7 In any $\triangle ABC$, prove that:

$$\frac{\cos A}{b \cos C + c \cos B} + \frac{\cos B}{c \cos A + a \cos C} + \frac{\cos C}{a \cos B + b \cos A} = \frac{a^2 + b^2 + c^2}{2abc}$$

SOLUTION We have,

$$\begin{aligned}
 \text{LHS} &= \frac{\cos A}{b \cos C + c \cos B} + \frac{\cos B}{c \cos A + a \cos C} + \frac{\cos C}{a \cos B + b \cos A} \\
 &= \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} \\
 &= \frac{b^2 + c^2 - a^2}{2abc} + \frac{c^2 + a^2 - b^2}{2abc} + \frac{a^2 + b^2 - c^2}{2abc} = \frac{a^2 + b^2 + c^2}{2abc} = \text{RHS}
 \end{aligned}$$

EXAMPLE 8 In a triangle ABC, if $\cos A = \frac{\sin B}{2 \sin C}$, show that the triangle is isosceles.

SOLUTION Let $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$. Then, $\sin A = ka$, $\sin B = kb$, $\sin C = kc$.

$$\text{Now, } \cos A = \frac{\sin B}{2 \sin C}$$

$$\Rightarrow 2 \cos A \sin C = \sin B$$

$$\Rightarrow 2 \left(\frac{b^2 + c^2 - a^2}{2bc} \right) kc = kb$$

$$\Rightarrow b^2 + c^2 - a^2 = b^2$$

$$\Rightarrow c^2 = a^2 \Rightarrow c = a$$

$$\Rightarrow \triangle ABC \text{ is isosceles.}$$

EXAMPLE 9 If in a triangle ABC, $\frac{2 \cos A}{a} + \frac{\cos B}{b} + \frac{2 \cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$, then prove that the triangle is right angled.

SOLUTION We have,

$$\frac{2 \cos A}{a} + \frac{\cos B}{b} + \frac{2 \cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$$

$$\Rightarrow 2 \left(\frac{b^2 + c^2 - a^2}{2abc} \right) + \left(\frac{c^2 + a^2 - b^2}{2abc} \right) + 2 \left(\frac{a^2 + b^2 - c^2}{2abc} \right) = \frac{a}{bc} + \frac{b}{ca}$$

$$\Rightarrow 2(b^2 + c^2 - a^2) + (c^2 + a^2 - b^2) + 2(a^2 + b^2 - c^2) = 2a^2 + 2b^2$$

$$\Rightarrow b^2 + c^2 = a^2$$

$$\Rightarrow \triangle ABC \text{ is a right angled triangle}$$

Type III ON FINDING THE AREA OF A TRIANGLE WHEN ITS PARTS ARE GIVEN

EXAMPLE 10 Find the area of a triangle ABC in which $\angle A = 60^\circ$, $b = 4 \text{ cm}$ and $c = \sqrt{3} \text{ cm}$.

SOLUTION The area Δ of triangle ABC is given by

$$\Delta = \frac{1}{2} bc \sin A = \frac{1}{2} \times 4 \sqrt{3} \times \sin 60^\circ = 2\sqrt{3} \times \frac{\sqrt{3}}{2} = 3 \text{ sq. cm.}$$

EXAMPLE 11 In any triangle ABC, prove that: $\Delta = \frac{b^2 + c^2 - a^2}{4 \cot A}$.

SOLUTION We have,

$$\begin{aligned} \text{RHS} &= \frac{b^2 + c^2 - a^2}{4 \cot A} = \frac{b^2 + c^2 - a^2}{4 \cos A} \sin A = \frac{b^2 + c^2 - a^2}{4(b^2 + c^2 - a^2)} \times 2bc \sin A \\ &= \frac{1}{2} bc \sin A = \Delta = \text{LHS} \end{aligned}$$

EXAMPLE 12 In any $\triangle ABC$, prove that: $\Delta = \frac{a^2 - b^2}{2} \cdot \frac{\sin A \sin B}{\sin(A - B)}$

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

$$\text{RHS} = \frac{a^2 - b^2}{2} \times \frac{\sin A \sin B}{\sin(A - B)} = \frac{k^2 \sin^2 A - k^2 \sin^2 B}{2} \times \frac{\sin A \sin B}{\sin(A - B)}$$

$$\begin{aligned}
 &= \frac{k^2}{2} (\sin^2 A - \sin^2 B) \times \frac{\sin A \sin B}{\sin(A-B)} = \frac{k^2}{2} \sin(A+B) \sin(A-B) \frac{\sin A \sin B}{\sin(A-B)} \\
 &= \frac{1}{2} k^2 \sin(A+B) \sin A \sin B = \frac{1}{2} (k \sin A) (k \sin B) \sin(\pi - C) \\
 &= \frac{1}{2} ab \sin C = \Delta = \text{LHS}.
 \end{aligned}$$

EXAMPLE 13 In any $\triangle ABC$, prove that: $a \cos A + b \cos B + c \cos C = \frac{8\Delta^2}{abc}$. [NCERT]

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

$$\begin{aligned}
 \text{Now, } a \cos A + b \cos B + c \cos C &= k \sin A \cos A + k \sin B \cos B + k \sin C \cos C \\
 &= \frac{k}{2} (\sin 2A + \sin 2B + \sin 2C) \\
 &= \frac{k}{2} (4 \sin A \sin B \sin C) = 2k \sin A \sin B \sin C = 2a \sin B \sin C \\
 &= 2a \times \frac{2\Delta}{ac} \times \frac{2\Delta}{ab} \quad \left[\because \Delta = \frac{1}{2} ab \sin C = \frac{1}{2} ac \sin B \therefore \sin B = \frac{2\Delta}{ac}, \sin C = \frac{2\Delta}{ab} \right] \\
 &= \frac{8\Delta^2}{abc} = \text{RHS}
 \end{aligned}$$

EXAMPLE 14 Two ships leave a port at the same time. One goes 24 km per hour in the direction $N 45^\circ E$ and other travels 32 km per hour in the direction $S 75^\circ E$. Find the distance between the ships at the end of 3 hours. [NCERT]

SOLUTION Let P and Q be the positions of two ships at the end of 3 hours. Then,

$$OP = 3 \times 24 = 72 \text{ km and } OQ = 3 \times 32 = 96 \text{ km}$$

Using cosine formula in $\triangle OPQ$, we get

$$\begin{aligned}
 PQ^2 &= OP^2 + OQ^2 - 2 OP \times OQ \cos 60^\circ \\
 \Rightarrow PQ^2 &= 72^2 + 96^2 - 2 \times 72 \times 96 \times \frac{1}{2} \\
 \Rightarrow PQ^2 &= 5184 + 9216 - 6912 = 7488 \\
 \Rightarrow PQ &= \sqrt{7488} \text{ km} = 86.533 \text{ km}
 \end{aligned}$$

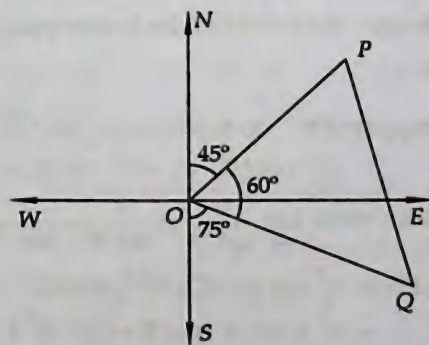


Fig. 10.13

EXAMPLE 15 Two boats leave a place at the same time. One travels 56 km in the direction $N 50^\circ E$, while other travels 48 km in the direction $S 80^\circ E$. What is the distance between the two positions of the boats?

SOLUTION Let A and B be the position of the boats such that $AB = x$.

$$\text{Clearly, } \angle AOB = 180^\circ - (50^\circ + 80^\circ) = 50^\circ$$

Using cosine formula, we have

$$AB^2 = OA^2 + OB^2 - 2 OA OB \cos \angle AOB$$

$$\Rightarrow x^2 = (56)^2 + (48)^2 - 2 \times 56 \times 48 \cos 50^\circ$$

$$\Rightarrow x^2 = 3136 + 2304 - 2 \times 56 \times 48 \times 0.6428$$

$$\Rightarrow x^2 = 5440 - 3455.69 = 1984.31$$

$$\Rightarrow x = \sqrt{1984.31} = 44.54 \text{ m}$$

Hence, the distance between the boats is 44.54 km.

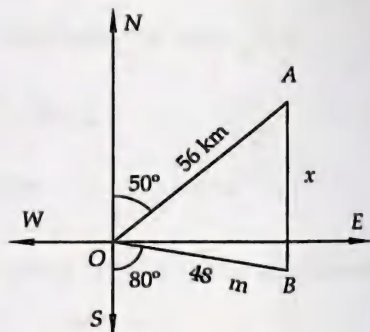


Fig. 10.14

EXAMPLE 16 A lamp-post is situated at the middle point M of the side AC of a triangular plot ABC with $BC = 7$ m, $CA = 8$ m and $AB = 9$ m. Lamp-post subtends an angle of 15° at the point B . Determine the height of the lamp-post. [NCERT]

SOLUTION Using cosine formula in ΔABC , we get

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 64 - 81}{2 \times 7 \times 8} = \frac{32}{112} = \frac{2}{7}$$

Using cosine formula in ΔBMC , we get

$$BM^2 = BC^2 + CM^2 - 2BC CM \cos C$$

$$\Rightarrow BM^2 = 49 + 16 - 2 \times 7 \times 4 \times \frac{2}{7} \quad \left[\because CM = \frac{1}{2} AC = 4 \right]$$

$$\Rightarrow BM^2 = 49 \Rightarrow BM = 7$$

In right triangle BMP , we have

$$\tan 15^\circ = \frac{PM}{BM} \Rightarrow \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{PM}{7} \Rightarrow PM = 7 \left(\frac{\sqrt{3}-1}{\sqrt{3}+1} \right) = 7(2-\sqrt{3}) \text{ m}$$

Hence, the height of the Lamp-post is $7(2-\sqrt{3})$ m.

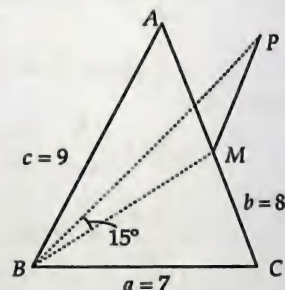


Fig. 10.15

LEVEL-2

EXAMPLE 17 In any triangle ABC , prove that:

$$a^3 \cos(B-C) + b^3 \cos(C-A) + c^3 \cos(A-B) = 3abc$$

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$

$$\text{LHS} = a^3 \cos(B-C) + b^3 \cos(C-A) + c^3 \cos(A-B)$$

$$= a^2 k \sin A \cos(B-C) + b^2 k \sin B \cos(C-A) + c^2 k \sin C \cos(A-B)$$

$$= \frac{k}{2} \left[a^2 \{ 2 \sin A \cos(B-C) \} + b^2 \{ 2 \sin B \cos(C-A) \} + c^2 \{ 2 \sin C \cos(A-B) \} \right]$$

$$= \frac{k}{2} \left[a^2 \{ 2 \sin(B+C) \cos(B-C) \} + b^2 \{ 2 \sin(C+A) \cos(C-A) \} \right.$$

$$\left. + c^2 \{ 2 \sin(A+B) \cos(A-B) \} \right]$$

$$= \frac{k}{2} \left[a^2 (\sin 2B + \sin 2C) + b^2 (\sin 2C + \sin 2A) + c^2 (\sin 2A + \sin 2B) \right]$$

$$\begin{aligned}
&= \frac{k}{2} \left[2a^2 (\sin B \cos B + \sin C \cos C) + 2b^2 (\sin C \cos C + \sin A \cos A) \right. \\
&\quad \left. + 2c^2 (\sin A \cos A + \sin B \cos B) \right] \\
&= \left[a^2 (k \sin B \cos B + k \sin C \cos C) + b^2 (k \sin C \cos C + k \sin A \cos A) \right. \\
&\quad \left. + c^2 (k \sin A \cos A + k \sin B \cos B) \right] \\
&= \left[a^2 (b \cos B + c \cos C) + b^2 (c \cos C + a \cos A) + c^2 (a \cos A + b \cos B) \right] \\
&\quad [\because k \sin A = a, k \sin B = b, k \sin C = c] \\
&= ab (a \cos B + b \cos A) + bc (b \cos C + c \cos B) + ca (a \cos C + c \cos A) \\
&= abc + bca + cab = 3abc
\end{aligned}$$

EXAMPLE 18 With usual notations, if in a triangle ABC $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$, then prove that:

$$\frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}$$

SOLUTION Let $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13} = \lambda$ (say) Then, $b+c=11\lambda$, $c+a=12\lambda$, $a+b=13\lambda$

$$\therefore (b+c+c+a+a+b) = 11\lambda + 12\lambda + 13\lambda \Rightarrow 2(a+b+c) = 36\lambda \Rightarrow a+b+c = 18\lambda$$

$$\text{Now, } b+c = 11\lambda \text{ and } a+b+c = 18\lambda \Rightarrow a = 7\lambda$$

$$c+a = 12\lambda \text{ and } a+b+c = 18\lambda \Rightarrow b = 6\lambda$$

$$a+b = 13\lambda \text{ and } a+b+c = 18\lambda \Rightarrow c = 5\lambda$$

$$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{36\lambda^2 + 25\lambda^2 - 49\lambda^2}{60\lambda^2} = \frac{12}{60} = \frac{1}{5}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ac} = \frac{25\lambda^2 + 49\lambda^2 - 36\lambda^2}{70\lambda^2} = \frac{38}{70} = \frac{19}{35}$$

$$\text{and, } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49\lambda^2 + 36\lambda^2 - 25\lambda^2}{84\lambda^2} = \frac{60}{84} = \frac{5}{7}$$

$$\therefore \cos A : \cos B : \cos C = \frac{1}{5} : \frac{19}{35} : \frac{5}{7} = 7 : 19 : 25$$

EXAMPLE 19 If a^2, b^2, c^2 are in A.P., prove that $\cot A, \cot B, \cot C$ are in A.P.

SOLUTION Let $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$. Then, $\sin A = ka$, $\sin B = kb$, $\sin C = kc$.

Now, $\cot A, \cot B, \cot C$ will be in A.P.

$$\Leftrightarrow 2 \cot B = \cot A + \cot C$$

$$\Leftrightarrow \frac{2 \cos B}{\sin B} = \frac{\cos A}{\sin A} + \frac{\cos C}{\sin C}$$

$$\Leftrightarrow \frac{2 \cos B}{kb} = \frac{\cos A}{ka} + \frac{\cos C}{kc}$$

$$\Leftrightarrow 2 \left(\frac{a^2 + c^2 - b^2}{2abc} \right) = \left(\frac{b^2 + c^2 - a^2}{2abc} \right) + \left(\frac{a^2 + b^2 - c^2}{2abc} \right)$$

$$\Leftrightarrow 2(a^2 + c^2 - b^2) = (b^2 + c^2 - a^2) + (a^2 + b^2 - c^2) \Leftrightarrow a^2 + c^2 = 2b^2 \Leftrightarrow a^2, b^2, c^2 \text{ are in A.P.}$$

ALITER It is given that a^2, b^2, c^2 are in A.P.

$$\therefore -2a^2, -2b^2, -2c^2 \text{ are in A.P.}$$

$$\Rightarrow (a^2 + b^2 + c^2) - 2a^2, (a^2 + b^2 + c^2) - 2b^2, (a^2 + b^2 + c^2) - 2c^2 \text{ are in A.P.}$$

$$\Rightarrow b^2 + c^2 - a^2, c^2 + a^2 - b^2, a^2 + b^2 - c^2 \text{ are in A.P.}$$

$$\Rightarrow \frac{b^2 + c^2 - a^2}{2abc}, \frac{c^2 + a^2 - b^2}{2abc}, \frac{a^2 + b^2 - c^2}{2abc} \text{ are in A.P.}$$

$$\Rightarrow \frac{1}{a} \left(\frac{b^2 + c^2 - a^2}{2bc} \right), \frac{1}{b} \left(\frac{c^2 + a^2 - b^2}{2ac} \right), \frac{1}{c} \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \text{ are in A.P.}$$

$$\Rightarrow \frac{\cos A}{k \sin A}, \frac{\cos B}{k \sin B}, \frac{\cos C}{k \sin C} \text{ are in A.P.} \quad [\because a = k \sin A, b = k \sin B, c = k \sin C]$$

$$\Rightarrow \cot A, \cot B, \cot C \text{ are in A.P.}$$

EXAMPLE 20 If in a triangle ABC, $\cos A + 2 \cos B + \cos C = 2$ prove that the sides of the triangle are in A.P.

SOLUTION We have,

$$\cos A + 2 \cos B + \cos C = 2$$

$$\Rightarrow \cos A + \cos C = 2 - 2 \cos B$$

$$\Rightarrow \cos A + \cos C = 2(1 - \cos B)$$

$$\Rightarrow 2 \cos \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right) = 2 \left(2 \sin^2 \frac{B}{2} \right)$$

$$\Rightarrow 2 \sin \frac{B}{2} \cos \left(\frac{A-C}{2} \right) = 4 \sin^2 \frac{B}{2} \quad \left[\because \cos \frac{A+C}{2} = \cos \left(\frac{\pi}{2} - \frac{B}{2} \right) = \sin \frac{B}{2} \right]$$

$$\Rightarrow \cos \left(\frac{A-C}{2} \right) = 2 \sin \frac{B}{2} \quad \left[\because 2 \sin \frac{B}{2} \neq 0 \right]$$

$$\Rightarrow 2 \cos \frac{B}{2} \cos \left(\frac{A-C}{2} \right) = 4 \sin \frac{B}{2} \cos \frac{B}{2} \quad \left[\text{Multiplying both sides by } 2 \cos \frac{B}{2} \right]$$

$$\Rightarrow 2 \sin \frac{A+C}{2} \cos \frac{A-C}{2} = 2 \left(2 \sin \frac{B}{2} \cos \frac{B}{2} \right) \quad \left[\because \cos \frac{B}{2} = \sin \frac{A+C}{2} \right]$$

$$\Rightarrow \sin A + \sin C = 2 \sin B$$

$$\Rightarrow ka + kc = 2kb \quad \left[\because \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k \right]$$

$$\Rightarrow a + c = 2b \Rightarrow a, b, c \text{ are in A.P.}$$

EXAMPLE 21 In a triangle ABC, $\angle C = 60^\circ$, then prove that : $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$.

SOLUTION We have, $\angle C = 60^\circ$

$$\Rightarrow \cos C = \frac{1}{2} \Rightarrow \frac{a^2 + b^2 - c^2}{2ab} = \frac{1}{2} \Rightarrow a^2 + b^2 - c^2 = ab \Rightarrow a^2 + b^2 - ab = c^2 \quad \dots(i)$$

$$\text{Now,} \quad \frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$$

$$\text{if} \quad \frac{a+b+2c}{(a+c)(b+c)} = \frac{3}{a+b+c}$$

$$\text{i.e. if} \quad (a+b+2c)(a+b+c) = 3(a+c)(b+c)$$

$$\text{i.e. if} \quad (a+b)^2 + 2c^2 + 3c(a+b) = 3(ab+ac+bc+c^2)$$

$$\text{i.e. if} \quad a^2 + b^2 + 2ab + 2c^2 + 3ac + 3bc = 3ab + 3ac + 3bc + 3c^2$$

$$\text{i.e. if} \quad a^2 + b^2 - ab = c^2, \text{ which is given}$$

[see (i)]

EXAMPLE 22 Two trees, A and B are on the same side of a river. From a point C in the river the distance of trees A and B are 250 m and 300 m respectively. If the angle C is 45° , find the distance between the trees (Use $\sqrt{2} = 1.44$).

SOLUTION Using cosine formula in $\triangle ABC$, we have

$$AB^2 = AC^2 + BC^2 - 2AC \cdot BC \cos \frac{\pi}{4}$$

$$\Rightarrow AB = \sqrt{(250)^2 + (300)^2 - 2 \times 250 \times 300 \times \frac{1}{\sqrt{2}}}$$

$$\Rightarrow AB = \sqrt{62500 + 90000 - 75000\sqrt{2}}$$

$$\Rightarrow AB = \sqrt{152500 - 75000 \times 1.44}$$

$$\Rightarrow AB = \sqrt{152500 - 108000} = \sqrt{44500} = 210.95 \text{ m}$$

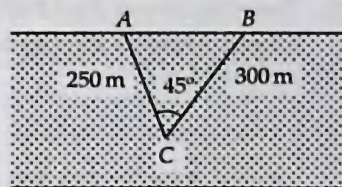


Fig. 10.16

EXERCISE 10.2

LEVEL-1

In any $\triangle ABC$, prove the following : (1-13)

1. In a $\triangle ABC$, if $a = 5$, $b = 6$ and $C = 60^\circ$, show that its area is $\frac{15\sqrt{3}}{2}$ sq. units.
2. In a $\triangle ABC$, if $a = \sqrt{2}$, $b = \sqrt{3}$ and $c = \sqrt{5}$, show that its area is $\frac{1}{2}\sqrt{6}$ sq. units.
3. The sides of a triangle are $a = 5$, $b = 6$ and $c = 8$, show that: $8 \cos A + 16 \cos B + 4 \cos C = 17$.
4. In a $\triangle ABC$, if $a = 18$, $b = 24$, $c = 30$, find $\cos A$, $\cos B$ and $\cos C$.
5. $b(c \cos A - a \cos C) = c^2 - a^2$
6. $c(a \cos B - b \cos A) = a^2 - b^2$
7. $2(bc \cos A + ca \cos B + ab \cos C) = a^2 + b^2 + c^2$
8. $(c^2 - a^2 + b^2) \tan A = (a^2 - b^2 + c^2) \tan B = (b^2 - c^2 + a^2) \tan C$
9. $\frac{c - b \cos A}{b - c \cos A} = \frac{\cos B}{\cos C}$
10. $a(\cos B + \cos C - 1) + b(\cos C + \cos A - 1) + c(\cos A + \cos B - 1) = 0$
11. $a \cos A + b \cos B + c \cos C = 2b \sin A \sin C$
12. $a^2 = (b + c)^2 - 4bc \cos^2 \frac{A}{2}$
13. $4 \left(bc \cos^2 \frac{A}{2} + ca \cos^2 \frac{B}{2} + ab \cos^2 \frac{C}{2} \right) = (a + b + c)^2$

LEVEL-2

14. In a $\triangle ABC$, prove that

$$\sin^3 A \cos(B - C) + \sin^3 B \cos(C - A) + \sin^3 C \cos(A - B) = 3 \sin A \sin B \sin C$$

15. In any $\triangle ABC$, $\frac{b+c}{12} = \frac{c+a}{13} = \frac{a+b}{15}$, then prove that $\frac{\cos A}{2} = \frac{\cos B}{7} = \frac{\cos C}{11}$.

16. In a $\triangle ABC$, if $\angle B = 60^\circ$, prove that $(a + b + c)(a - b + c) = 3ca$

17. If in a $\triangle ABC$, $\cos^2 A + \cos^2 B + \cos^2 C = 1$, prove that the triangle is right angled.

18. In a ΔABC , if $\cos C = \frac{\sin A}{2 \sin B}$, prove that the triangle is isosceles.
19. Two ships leave a port at the same time. One goes 24 km/hr in the direction $N 38^\circ E$ and other travels 32 km/hr in the direction $S 52^\circ E$. Find the distance between the ships at the end of 3 hrs.

ANSWERS

$$4. \cos A = \frac{4}{5}, \cos B = \frac{3}{5}, \cos C = 0 \quad 19. 120 \text{ km}$$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

- Find the area of the triangle ΔABC in which $a = 1$, $b = 2$ and $\angle C = 60^\circ$.
- In a ΔABC , if $b = \sqrt{3}$, $c = 1$ and $\angle A = 30^\circ$, find a .
- In a ΔABC , if $\cos A = \frac{\sin B}{2 \sin C}$, then show that $c = a$.
- In a ΔABC , if $b = 20$, $c = 21$ and $\sin A = \frac{3}{5}$, find a .
- In a ΔABC , if $\sin A$ and $\sin B$ are the roots of the equation $c^2 x^2 - c(a+b)x + ab = 0$, then find $\angle C$.
- In ΔABC , if $a = 8$, $b = 10$, $c = 12$ and $C = \lambda A$, find the value of λ .
- If the sides of a triangle are proportional to 2, $\sqrt{6}$ and $\sqrt{3} - 1$, find the measure of its greatest angle.
- If in a ΔABC , $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$, then find the measures of angles A , B , C .
- In any triangle ABC , find the value of $a \sin(B-C) + b \sin(C-A) + c \sin(A-B)$.
- In any ΔABC , find the value of $\Sigma a(\sin B - \sin C)$.

ANSWERS

1. $\sqrt{3}$ sq. units 2. 1 4. 13 5. 90° 6. 2 7. 120° 8. $A = B = C = 60^\circ$ 9. 0 10. 0

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

- In any ΔABC , $\sum a^2(\sin B - \sin C) =$
 (a) $a^2 + b^2 + c^2$ (b) a^2 (c) b^2 (d) 0
- In a ΔABC , if $a = 2$, $\angle B = 60^\circ$ and $\angle C = 75^\circ$, then $b =$
 (a) $\sqrt{3}$ (b) $\sqrt{6}$ (c) $\sqrt{9}$ (d) $1 + \sqrt{2}$
- If the sides of a triangle are in the ratio $1 : \sqrt{3} : 2$, then the measure of its greatest angle is
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$
- In any ΔABC , $2(bc \cos A + ca \cos B + ab \cos C) =$
 (a) abc (b) $a + b + c$ (c) $a^2 + b^2 + c^2$ (d) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$

5. In a triangle ABC , $a = 4$, $b = 3$, $\angle A = 60^\circ$ then c is a root of the equation
(a) $c^2 - 3c - 7 = 0$ (b) $c^2 + 3c + 7 = 0$ (c) $c^2 - 3c + 7 = 0$ (d) $c^2 + 3c - 7 = 0$
6. In a $\triangle ABC$, if $(c + a + b)(a + b - c) = ab$, then the measure of angle C is
(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{2}$
7. In any $\triangle ABC$, the value of $2ac \sin\left(\frac{A - B + C}{2}\right)$ is
(a) $a^2 + b^2 - c^2$ (b) $c^2 + a^2 - b^2$ (c) $b^2 - c^2 - a^2$ (d) $c^2 - a^2 - b^2$
8. In any $\triangle ABC$, $a(b \cos C - c \cos B) =$
(a) a^2 (b) $b^2 - c^2$ (c) 0 (d) $b^2 + c^2$

ANSWERS

1. (d) 2. (b) 3. (c) 4. (c) 5. (a) 6. (a) 7. (c) 8. (c)

TRIGONOMETRIC EQUATIONS

11.1 SOME DEFINITIONS

TRIGONOMETRIC EQUATIONS The equations containing trigonometric functions of unknown angles are known as trigonometric equations.

$\cos \theta = \frac{1}{2}$, $\sin \theta = 0$, $\tan \theta = \sqrt{3}$ etc. are trigonometric equations.

SOLUTION OF A TRIGONOMETRIC EQUATION A solution of a trigonometric equation is the value of the unknown angle that satisfies the equation.

Consider the equation $\sin \theta = \frac{1}{2}$. This equation is clearly satisfied by $\theta = \frac{\pi}{6}$, $\frac{5\pi}{6}$ etc. So, these are its solutions.

Solving an equation means to find the set of all values of the unknown angle which satisfy the given equation.

Consider the equation $2 \cos \theta + 1 = 0$ or $\cos \theta = -1/2$. This equation is clearly satisfied by $\theta = \frac{2\pi}{3}$, $\frac{4\pi}{3}$ etc.

Since the trigonometric functions are periodic. Therefore, if a trigonometric equation has a solution, it will have infinitely many solutions. For example, $\theta = \frac{2\pi}{3}$, $2\pi \pm \frac{2\pi}{3}$, $4\pi \pm \frac{2\pi}{3}$, are solutions of $2 \cos \theta + 1 = 0$. These solutions can be put together in compact form as $2n\pi \pm \frac{2\pi}{3}$,

where n is an integer. This solution is known as the general solution.

Thus, a solution generalised by means of periodicity is known as the general solution.

It also follows from the above discussion that solving an equation means to find its general solution.

11.2 GENERAL SOLUTIONS OF TRIGONOMETRIC EQUATIONS

In this section, we shall obtain the general solutions of the trigonometric equations $\sin \theta = 0$, $\cos \theta = 0$, $\tan \theta = 0$ and $\cot \theta = 0$.

THEOREM 1 Prove that the general solution of $\sin \theta = 0$ is given by $\theta = n\pi$, $n \in \mathbb{Z}$.

PROOF In $\triangle OMP$, we obtain

$$\sin \theta = \frac{PM}{OP}$$

$$\therefore \sin \theta = 0$$

$$\Rightarrow \frac{PM}{OP} = 0$$

$$\Rightarrow PM = 0$$

$$\Rightarrow OP \text{ coincides with } OX \text{ or } OX'$$

$$\Rightarrow \theta = 0, \pi, 2\pi, \dots, -\pi, -2\pi, -3\pi, \dots$$

$$\Rightarrow \theta = n\pi, n \in \mathbb{Z}.$$

Hence, $\theta = n\pi$, $n \in \mathbb{Z}$ is the general solution of $\sin \theta = 0$.

Q.E.D.

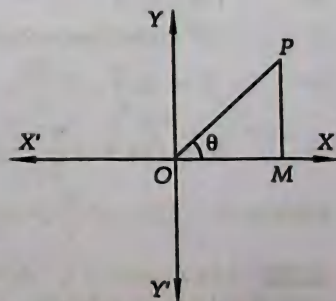


Fig. 11.1

THEOREM 2 Prove that the general solution of $\tan \theta = 0$ is $\theta = n\pi$, $n \in \mathbb{Z}$.

PROOF By definition,

$$\tan \theta = \frac{PM}{OM}$$

[See Fig. 11.1]

$$\therefore \tan \theta = 0$$

$$\Rightarrow \frac{PM}{OM} = 0$$

X or, OX'

$\pi, -2\pi, \dots$

ral solution of $\sin \theta = 0$.

Q.E.D.

ral solution of $\cos \theta = 0$ is $\theta = (2n+1)\frac{\pi}{2}$, $n \in \mathbb{Z}$.

[See Fig. 11.1]

$$\begin{aligned} & \text{OY'} \\ & \dots \\ & 2 \\ & 2, n \in \mathbb{Z}. \end{aligned}$$

Hence, the general solution of $\cos \theta = 0$ is $\theta = (2n+1)\frac{\pi}{2}$, $n \in \mathbb{Z}$.

Q.E.D.

THEOREM 4 Prove that the general solution of $\cot \theta = 0$ is $\theta = (2n+1)\frac{\pi}{2}$, $n \in \mathbb{Z}$.

PROOF By definition,

$$\cot \theta = \frac{OM}{PM}$$

[See Fig. 11.1]

$$\therefore \cot \theta = 0$$

$$\Rightarrow \frac{OM}{PM} = 0$$

$$\Rightarrow OM = 0$$

$$\Rightarrow OP \text{ coincides with } OY \text{ or, } OY'$$

$$\Rightarrow \theta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$$

$$\Rightarrow \theta = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}.$$

Hence, $\theta = (2n+1)\frac{\pi}{2}$, $n \in \mathbb{Z}$ is the general solution of $\cot \theta = 0$.

Q.E.D.

NOTE Since $\sec \theta \geq 1$, or $\sec \theta \leq -1$, therefore $\sec \theta = 0$ does not have any solution. Similarly, $\operatorname{cosec} \theta = 0$ has no solution.

ILLUSTRATIVE EXAMPLES**LEVEL-1****EXAMPLE 1** Find the general solutions of the following equations:

$$(i) \sin 2\theta = 0 \quad (ii) \sin \frac{3\theta}{2} = 0 \quad (iii) \sin^2 2\theta = 0$$

SOLUTION (i) We have,

$$\sin 2\theta = 0$$

$$\Rightarrow 2\theta = n\pi, \text{ where } n \in \mathbb{Z}$$

$$[\because \sin \theta = 0 \Rightarrow \theta = n\pi]$$

$$\Rightarrow \theta = \frac{n\pi}{2}, n \in \mathbb{Z}.$$

$$(ii) \sin \frac{3\theta}{2} = 0$$

$$\Rightarrow \frac{3\theta}{2} = n\pi, n \in \mathbb{Z}$$

$$[\sin \theta = 0 \Rightarrow \theta = n\pi]$$

$$\Rightarrow \theta = \frac{2n\pi}{3}, n \in \mathbb{Z}.$$

$$(iii) \sin^2 2\theta = 0 \Rightarrow \sin 2\theta = 0 \Rightarrow 2\theta = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{n\pi}{2}, n \in \mathbb{Z}.$$

EXAMPLE 2 Find the general solutions of the following equations:

$$(i) \cos 3\theta = 0 \quad (ii) \cos \frac{3\theta}{2} = 0 \quad (iii) \cos^2 3\theta = 0$$

SOLUTION We know that the general solution of the equation $\cos \theta = 0$ is $\theta = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$.

Therefore,

$$(i) \cos 3\theta = 0 \Rightarrow 3\theta = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \Rightarrow \theta = (2n+1)\frac{\pi}{6}, n \in \mathbb{Z}$$

$$(ii) \cos \frac{3\theta}{2} = 0 \Rightarrow \frac{3\theta}{2} = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \Rightarrow \theta = (2n+1)\frac{\pi}{3}, n \in \mathbb{Z}.$$

$$(iii) \cos^2 3\theta = 0 \Rightarrow \cos 3\theta = 0 \Rightarrow 3\theta = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \Rightarrow \theta = (2n+1)\frac{\pi}{6}, n \in \mathbb{Z}.$$

EXAMPLE 3 Find the general solutions of the following equations:

$$(i) \tan 2\theta = 0 \quad (ii) \tan \frac{\theta}{2} = 0 \quad (iii) \tan \frac{3\theta}{4} = 0$$

SOLUTION We know that the general solution of the equation $\tan \theta = 0$ is $\theta = n\pi, n \in \mathbb{Z}$.

Therefore,

$$(i) \tan 2\theta = 0 \Rightarrow 2\theta = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{n\pi}{2}, n \in \mathbb{Z}$$

$$(ii) \tan \frac{\theta}{2} = 0 \Rightarrow \frac{\theta}{2} = n\pi, n \in \mathbb{Z} \Rightarrow \theta = 2n\pi, n \in \mathbb{Z}$$

$$(iii) \tan \frac{3\theta}{4} = 0 \Rightarrow \frac{3\theta}{4} = n\pi, n \in \mathbb{Z} \Rightarrow \theta = \frac{4n\pi}{3}, n \in \mathbb{Z}$$

THEOREM 5 Prove that the general solution of $\sin \theta = \sin \alpha$ is given by: $\theta = n\pi + (-1)^n \alpha, n \in \mathbb{Z}$.**PROOF** We have,

$$\sin \theta = \sin \alpha$$

$$\Leftrightarrow \sin \theta - \sin \alpha = 0$$

$$\Leftrightarrow 2 \sin \left(\frac{\theta - \alpha}{2} \right) \cos \left(\frac{\theta + \alpha}{2} \right) = 0$$